Measure Theoretic Probability
Lesson 12
Review of Independence: Generated σ-Fields, Uniqueness of Measures, and an Event vs. a σ-Field

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Outline

1. Recap and an Apparent Contradiction

2. Uniqueness of Measures

3. An Event Is Independent of a Generated σ-Field

4. Summary and Exercises

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Recap from What Lesson 10

Theorem (Lesson 10)

If $\mathcal{A}_1, \mathcal{A}_2, \ldots$ are independent π-**systems**, then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \ldots$ are independent.

Without the π-system assumption it fails

Independent classes need not generate independent σ-fields. The counterexample used two fair coin tosses.

The plan

A new statement will look very similar to that counterexample. We must check carefully that there is **no contradiction**, and then prove the result we need for Kolmogorov's 0–1 law.

The Counterexample, with New Notation

Two fair coin tosses

$\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$. Let

$$A_1 = \{HT, TT\}, \qquad A_2 = \{HH, TT\}, \qquad A_3 = \{HH, HT\},$$

$$\mathcal{A}_1 = \{A_1\}, \qquad \mathcal{A}_2 = \{A_2, A_3\}.$$

The classes are independent

$\mathbb{P}(A_1 \cap A_2) = \mathbb{P}(\{TT\}) = \frac{1}{4} = \mathbb{P}(A_1)\mathbb{P}(A_2)$ and $\mathbb{P}(A_1 \cap A_3) = \mathbb{P}(\{HT\}) = \frac{1}{4} = \mathbb{P}(A_1)\mathbb{P}(A_3)$.
(Take away: $\mathbb{P}(A_2 \cap A_3) = \mathbb{P}(\{HH\}) = \frac{1}{4}$, so A_2 and A_3 are independent too.
This is not required for the classes to be independent.)

But the generated σ-fields are not independent

$A_1 \in \sigma(\mathcal{A}_1)$ and $A_1 = (A_2 \cup A_3) \setminus (A_2 \cap A_3) \in \sigma(\mathcal{A}_2)$. A set independent of itself needs $\mathbb{P}(A_1) \in \{0, 1\}$, but $\mathbb{P}(A_1) = \frac{1}{2}$.

A New Claim That Looks Similar

Claim 1

Let A_1, A_2, A_3 be (mutually) independent events. Then $\sigma(A_1), \sigma(A_2), \sigma(A_3)$ are independent σ-fields, where

$$\sigma(A_i) = \{\varnothing, A_i, A_i^c, \Omega\}.$$

Why this is not surprising

- Complements of independent events are independent, so picking one set from each four-element σ-field still factors (also for triple intersections).
- Each class $\{A_i\}$ has one set, hence is a π-system ($A_i \cap A_i = A_i$)

Claim 2 (more surprising)

$\sigma(A_1)$ is independent of $\sigma(A_2, A_3)$, although $\{A_2, A_3\}$ is in general **not** a π-system ($A_2 \cap A_3 \notin \{A_2, A_3\}$). In particular A_1 is independent of $\sigma(A_2, A_3)$.

Resolving the Apparent Contradiction

Coin example

A_1, A_2, A_3 are only **pairwise** independent. The triple intersection is

$$A_1 \cap A_2 \cap A_3 = \varnothing,$$

so

$$\mathbb{P}(A_1 \cap A_2 \cap A_3) = 0 \neq \tfrac{1}{8} = \mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3).$$

Not mutually independent.

Claim 2

A_1, A_2, A_3 are **mutually** independent.

The key difference

Independence of the classes $\{A_1\}$ and $\{A_2, A_3\}$ only tests **one set from each class**. It never tests $A_2 \cap A_3$ together with A_1. Mutual independence of the three sets does, and that is exactly what rescues the generated σ-fields.

Conclusion

No contradiction: Claim 2 assumes **more** independence than the coin example had. Pairwise independence is not enough; mutual independence of the individual sets is.

The Bigger Picture: An Array of Events

Statement

Let $\{A_{ij}\}$ be an array of events

$$\begin{array}{cccc} A_{11} & A_{12} & A_{13} & \cdots \\ A_{21} & A_{22} & A_{23} & \cdots \\ \vdots & \vdots & \vdots & \end{array}$$

that are all **mutually independent**. Let $\mathcal{F}_i = \sigma(A_{i1}, A_{i2}, \dots)$ be the σ-field generated by row i. Then $\mathcal{F}_1, \mathcal{F}_2, \dots$ are independent.

What we prove

A slightly weaker version with the same method, which is exactly what Kolmogorov's 0–1 law needs:

Theorem 2. If $A_1, A_2, \dots$ are independent, then A_1 is independent of $\sigma(A_2, A_3, \dots)$.

The general case follows by repeating the argument, but is more tedious.

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Theorem: Measures Agreeing on a π-System

Setting

$\Omega \neq \varnothing$; $\mathcal{P}$ a π-system of subsets of Ω; $\mathcal{F} = \sigma(\mathcal{P})$. Let μ_1, μ_2 be **finite** measures on $(\Omega, \mathcal{F})$ with

$$\mu_1(\Omega) = \mu_2(\Omega).$$

(Automatic for probability measures: both equal 1.)

Theorem

If $\mu_1(A) = \mu_2(A)$ for every $A \in \mathcal{P}$, then

$$\mu_1(A) = \mu_2(A) \quad \text{for every } A \in \sigma(\mathcal{P}) = \mathcal{F}.$$

In earlier lessons

In the outer-measure lessons we stated a uniqueness result for generating **fields**. Since a π-system is a weaker assumption than a field, this theorem contains it.

Proof Strategy and Definition of $\mathcal{D}$

Define the agreement class

$$\mathcal{D} = \{A \in \mathcal{F} : \ \mu_1(A) = \mu_2(A)\} \subseteq \mathcal{F}.$$

What we need

By hypothesis $\mathcal{P} \subseteq \mathcal{D}$. If $\mathcal{D}$ is a λ-system, then Dynkin's π-λ theorem ($\mathcal{P}$ a π-system, $\mathcal{D}$ a λ-system, $\mathcal{P} \subseteq \mathcal{D}$) gives

$$\mathcal{F} = \sigma(\mathcal{P}) \subseteq \mathcal{D} \subseteq \mathcal{F},$$

so $\mathcal{D} = \mathcal{F}$, i.e. the measures agree on all of $\mathcal{F}$.

Remaining task

Show that $\mathcal{D}$ is a λ-system: (i) $\Omega \in \mathcal{D}$, (ii) closed under complements, (iii) closed under countable disjoint unions.

Proof that $\mathcal{D}$ Is a λ-System

(i) $\Omega \in \mathcal{D}$

By assumption $\mu_1(\Omega) = \mu_2(\Omega)$.

(ii) Complements

Let $A \in \mathcal{D}$. Since A and A^c are disjoint with union Ω, finite additivity gives $\mu_i(A) + \mu_i(A^c) = \mu_i(\Omega)$. Because the measures are **finite** we may subtract:

$$\mu_1(A^c) = \mu_1(\Omega) - \mu_1(A) = \mu_2(\Omega) - \mu_2(A) = \mu_2(A^c).$$

So $A^c \in \mathcal{D}$.

(iii) Countable disjoint unions

Let $A_1, A_2, \cdots \in \mathcal{D}$ be pairwise disjoint. By countable additivity of each μ_i,

$$\mu_1\Big(\bigcup_n A_n\Big) = \sum_n \mu_1(A_n) = \sum_n \mu_2(A_n) = \mu_2\Big(\bigcup_n A_n\Big).$$

So $\bigcup_n A_n \in \mathcal{D}$. □

Where finiteness is used

Only in (ii), to subtract. For infinite measures the theorem can fail without extra hypotheses.

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Theorem 2 and the Plan

Theorem 2

Let $A_1, A_2, A_3, \cdots \in \mathcal{F}$ be mutually independent events (a countable list; independence of an infinite family means independence of all finite sub-families). Then A_1 is independent of $\sigma(A_2, A_3, \dots)$: for every $A \in \sigma(A_2, A_3, \dots)$,

$$\mathbb{P}(A \cap A_1) = \mathbb{P}(A)\,\mathbb{P}(A_1).$$

Strategy

1. Build a π-system $\mathcal{A}$ generating $\sigma(A_2, A_3, \dots)$ on which the product rule is easy.
2. Define a new probability measure $\tilde{\mathbb{P}}$ from A_1 and show it equals $\mathbb{P}$ on $\mathcal{A}$.
3. Apply the uniqueness theorem: $\tilde{\mathbb{P}} = \mathbb{P}$ on the whole σ-field.

Step 1: The π-System $\mathcal{A}$

Definition

Let $\mathcal{A}$ be the class of all finite intersections

$$E_2 \cap E_3 \cap \cdots \cap E_n, \qquad n \geq 2,$$

where each $E_k \in \{A_k,\ A_k^c,\ \varnothing,\ \Omega\}$.

Properties

- π-**system:** an intersection of two such sets is again a finite intersection of this form (pad shorter ones with Ω and intersect coordinatewise; e.g. $A_k \cap A_k^c = \varnothing$ is also allowed).
- $\sigma(\mathcal{A}) = \sigma(A_2, A_3, \dots)$. ($\supseteq$) Take $E_k = A_k$ and all other $E_j = \Omega$: each $A_k \in \mathcal{A}$. ($\subseteq$) Every set in $\mathcal{A}$ is built from the A_k by complements and finite intersections.

Product rule on $\mathcal{A}$

For $A = E_2 \cap \cdots \cap E_n \in \mathcal{A}$, mutual independence of $A_1, \dots, A_n$ (and of their complements) gives

$$\mathbb{P}(A \cap A_1) = \mathbb{P}(A_1) \prod_{k=2}^{n} \mathbb{P}(E_k) = \mathbb{P}(A_1)\,\mathbb{P}(A).$$

Step 2: A New Probability Measure

Definition (assume $\mathbb{P}(A_1) > 0$)

For $A \in \sigma(A_2, A_3, \dots)$ put

$$\tilde{\mathbb{P}}(A) = \frac{\mathbb{P}(A \cap A_1)}{\mathbb{P}(A_1)}.$$

$\tilde{\mathbb{P}}$ is a probability measure on $\sigma(A_2, A_3, \dots)$

- $\tilde{\mathbb{P}}(\varnothing) = 0$ and $\tilde{\mathbb{P}}(A) \geq 0$.
- $\tilde{\mathbb{P}}(\Omega) = \dfrac{\mathbb{P}(\Omega \cap A_1)}{\mathbb{P}(A_1)} = \dfrac{\mathbb{P}(A_1)}{\mathbb{P}(A_1)} = 1.$
- Countable additivity: if the B_m are disjoint, so are the $B_m \cap A_1$, and $\mathbb{P}$ is countably additive; divide by $\mathbb{P}(A_1)$.

Agreement on $\mathcal{A}$

By Step 1, for $A \in \mathcal{A}$: $\tilde{\mathbb{P}}(A) = \dfrac{\mathbb{P}(A)\mathbb{P}(A_1)}{\mathbb{P}(A_1)} = \mathbb{P}(A).$

The degenerate case

If $\mathbb{P}(A_1) = 0$, then $\mathbb{P}(A \cap A_1) \leq \mathbb{P}(A_1) = 0 = \mathbb{P}(A)\mathbb{P}(A_1)$ for every A, so the claim is trivial.

Step 3: Apply Uniqueness

Conclusion of the proof

$\mathbb{P}$ and $\tilde{\mathbb{P}}$ are probability measures on $\sigma(A_2, A_3, \dots) = \sigma(\mathcal{A})$ that agree on the π-system $\mathcal{A}$. By the uniqueness theorem they agree on all of $\sigma(A_2, A_3, \dots)$. So for every $A \in \sigma(A_2, A_3, \dots)$,

$$\mathbb{P}(A) = \tilde{\mathbb{P}}(A) = \frac{\mathbb{P}(A \cap A_1)}{\mathbb{P}(A_1)} \qquad \Longrightarrow \qquad \mathbb{P}(A \cap A_1) = \mathbb{P}(A_1)\,\mathbb{P}(A). \qquad \square$$

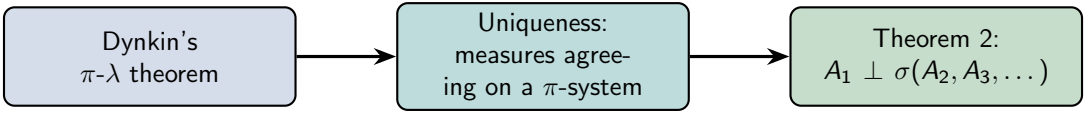
Reading the result

The single event A_1 (equivalently the class $\{A_1\}$) is independent of the whole σ-field generated by all the other events.

Why this is the result we need

Events that depend only on the "tail" $A_n, A_{n+1}, \dots$ are independent of $\sigma(A_1, \dots, A_{n-1})$. This is the engine behind Kolmogorov's 0–1 law. We will see this in future lessons

How the Pieces Fit



$\mathcal{D} = \{\mu_1 = \mu_2\}$
is a λ-system

$\tilde{\mathbb{P}}(\cdot) = \mathbb{P}(\cdot \cap A_1)/\mathbb{P}(A_1)$
and $\mathbb{P}$ agree on $\mathcal{A}$

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Summary

Key results

Concept / Result	Content
Lesson 10 theorem	independent π-systems $\Rightarrow$ independent generated σ-fields
Coin counterexample	A_1, A_2, A_3 only **pairwise** independent; $A_1 \in \sigma(\mathcal{A}_1) \cap \sigma(\mathcal{A}_2)$
Mutual independence	$\sigma(A_1)$, $\sigma(A_2)$, $\sigma(A_3)$ independent; $A_1 \perp \sigma(A_2, A_3)$; no contradiction
Uniqueness theorem	finite measures with $\mu_1(\Omega) = \mu_2(\Omega)$ agreeing on a π-system agree on $\sigma(\mathcal{P})$
Proof tool	$\{\mu_1 = \mu_2\}$ is a λ-system + Dynkin
Theorem 2	$A_1, A_2, \ldots$ independent $\Rightarrow A_1 \perp \sigma(A_2, A_3, \ldots)$

Exercises

1. In the coin example, verify directly that A_1, A_2, A_3 are pairwise independent but not mutually independent. Find a set in $\sigma(\mathcal{A}_1) \cap \sigma(\mathcal{A}_2)$ of probability $\frac{1}{2}$.
2. Show that the uniqueness theorem fails if $\mathcal{P}$ is not a π-system: find two probability measures on $\{1,2,3,4\}$ that agree on $\{\{1,2\},\{2,3\}\}$ but differ elsewhere.
3. Let μ_1, μ_2 be probability measures on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ with $\mu_1((-\infty, x]) = \mu_2((-\infty, x])$ for all $x \in \mathbb{R}$. Prove $\mu_1 = \mu_2$.
4. Verify that $\tilde{\mathbb{P}}(A) = \mathbb{P}(A \cap A_1)/\mathbb{P}(A_1)$ satisfies countable additivity.
5. Using Theorem 2, prove that if A_1, A_2, A_3, A_4 are independent, then $\sigma(A_1, A_2)$ is independent of $\sigma(A_3, A_4)$ by repeating the argument twice.

Outlook

Next lesson

Tail σ-fields (tail σ-algebras): for a sequence of events $A_1, A_2, \ldots$ look at what can be described using $A_n, A_{n+1}, \ldots$ for every n, so any finite number of initial terms can be dropped.

Goal: Kolmogorov's 0–1 law

A generalization of the Borel–Cantelli lemmas: for independent sequences, every tail event has probability 0 or 1. After that, we finally introduce **random variables**.